DNMS: DISTRIBUTED NETWORK MONITORING SYSTEM

Multi-Path ECMP Sweeping • High-Frequency UDP Telemetry • AI-Free Mathematical Link Fault Attribution

CORE RUNTIME: Go 1.23+ & Linux Raw Sockets
TOPOLOGY MODEL: 18-Node Campus Spine-Leaf
FAULT ATTRIBUTION: Deterministic Formula (Zero AI)
MISSION CONTROL: D3.js NOC + Wireshark Dock

SWIM GOSSIP	DEDICATED UDP PING	TRACEROUTE SWEEPER	REST & SSE GATEWAY	PROBE RESOLUTION	METRIC CONVERGENCE
Port 33434 UDP/TCP	Port 33435 UDP	Port 33436+ UDP	Port 12345 HTTP	100ms Cycle / 1.2ms RTT	<1.0s Fast Ring Reset

1. THE PROBLEM: ECMP BLIND SPOT

CRITICAL FLAW

Modern cloud datacenters and spine-leaf fabrics utilize **Equal-Cost Multi-Path (ECMP)** routing to distribute traffic across parallel links using a 5-tuple hash:
Hash = CRC32(SrcIP, DstIP, Proto, SrcPort, DstPort) % N

THE MULTI-PATH HASH POLARIZATION DILEMMA
[Traditional ICMP Ping / MTR]
• Static ports / no port numbers → Single static hash bucket.
• Follows **Path 1 (Spine 01)** [0% Loss, Healthy].
• Dashboard reports: **ALL GREEN** → *False Sense of Security!*
• **Path 2 (Spine 02)** silently drops 40% customer packets!

[DNMS Source-Port Sweeping]
• Iterates UDP source ports (33436–33450) → Triggers all hash buckets.
• Exposes both Path 1 and Path 2 simultaneously on topology canvas.

Silent Packet Drops: Gray failures on alternate spine links remain completely invisible to legacy ping tools.

SNMP Credential Overhead: Traditional NMS requires SSH/SNMP credentials on every switch—forbidden in zero-trust.

Prolonged MTTR: Engineers spend hours running manual traceroutes to isolate degrading physical cables.

2. DECENTRALIZED SWIM GOSSIP

ZERO-CONFIG

DNMS operates as an autonomous, self-healing peer-to-peer mesh using **HashiCorp Memberlist** implementing the SWIM gossip protocol:

No Central Coordinator: Zero ZooKeeper, Consul, or etcd dependencies. Eliminates single point of failure.

Scalable O(log N) Gossip: Nodes exchange heartbeats over UDP/TCP 33434 with bounded message overhead.

Indirect Peer Probing: Suspect nodes are probed through random third-party peers to prevent false-positives.

SWIM PROTOCOL FINITE STATE MACHINE
• **Direct Ping:** Node A sends ping to Node B over UDP 33434.
• **Indirect Ping:** If no ACK within 200ms, Node A requests 3 random peers to probe Node B.
• **Suspect State:** If indirect pings fail, Node B enters 'Suspect' state with an active refute timer.
• **Dead Declaration:** Node B marked dead and broadcast to mesh.

3. TARGET PERSONAS & ROLES

ENTERPRISE

Datacenter NOC Engineers: Monitor 10,000+ leaf-spine cables with real-time visual alerts and direct link fault context menus.

Cloud Platform SREs: Instantly isolate degrading links causing microservice timeouts without access to underlying cloud switches.

High-Frequency Trading (HFT): Detect microsecond latency jitter and standard deviation spikes ruins automated trading algos.

Enterprise Campus NetOps: Segmented subnet monitoring (Management, Eng, Finance, DMZ) with automated JSON topology export.

ENTERPRISE ZERO-CREDENTIAL COMPLIANCE
• Standard monitoring tools violate zero-trust security by demanding read/write switch credentials.
• **DNMS operates entirely at the transport edge:** infers link health mathematically from active route probes without logging into any switch.

4. AUTONOMOUS PIPELINE

END-TO-END

01 PEER DISCOVERY • Port 33434 UDP/TCP
SWIM gossip discovers cluster nodes and metadata.

02 ECMP SWEEPING • Port 33436+ UDP
Sweeps source ports; Linux raw sockets intercept TTL expiry.

03 UDP PING TELEMETRY • Port 33435 UDP
High-frequency binary UDP probes calculate RTT and loss.

04 ROLLING METRICS RING • 100 Samples / Route
container/ring maintains loss rate, jitter, and p50/p90/p99.

05 FAULT ATTRIBUTION • Mathematical Score
Correlates route loss with link membership: Loss(thru) - Loss(not).

06 MISSION CONTROL NOC • Port 12345 / SSE
D3.js force layout streams live mutations & Wireshark packets.

5. LINK FAULT ATTRIBUTION

DETERMINISTIC

DNMS isolates broken and degraded links using probabilistic cross-path loss correlation, eliminating router login requirements:

PROBABILISTIC FAULT LOCALIZATION EQUATION
Suspicion(e) = AvgLossThrough(e) – AvgLossNotThrough(e)
Where AvgLoss is sample-weighted across all traversing end-to-end paths

AvgLossThrough(e): Sample-weighted average loss rate of all discovered routes that traverse physical link e.

AvgLossNotThrough(e): Baseline average loss rate across all routes in the network that do NOT traverse link e.

Zero Hallucinations: Pure deterministic mathematics. Eliminates unreliable AI models and false alarms.

Fault Threshold (> 10%): Links exhibiting positive suspicion over 10% are badged with a bright red **FAULT SUSPECT** tag.

6. WIRESHARK DISSECTOR DOCK

REAL-TIME

An enterprise-grade packet inspection dock embedded directly in the Web NOC dashboard, displaying every network frame with millisecond timestamps:

LIVE PACKET STREAM INSPECTOR
• **Millisecond Timestamps:** High-resolution HH:MM:SS.mmm clock.
• **Multi-Protocol Dissection:** Decodes ICMP, TCP, UDP, SWIM, and BGP frames.
• **Dynamic Protocol Filter Bar:** Instant substring search across packet sources, destinations, and protocols.
• **Operational State Frames:** Captures LINK_DOWN, LINK_UP, and QOS_IMPAIR events in-stream.
• **Zero-Friction Inspection:** Toggle dock with W shortcut or full-screen zen focus mode.

7. ENTERPRISE FABRIC MODEL

18 NODES

DNMS deploys a full-scale 18-node Enterprise Campus Network Model with Core-Distribution-Access hierarchy and redundant ECMP paths:

Core Gateways: 2x Border Core Routers (core-gw-01, core-gw-02) terminating 40G trunks.

Next-Gen Firewalls: 2x Redundant Stateful Firewalls (sec-fw-01, sec-fw-02) filtering DMZ flows.

Distribution Tier: 2x Multi-Chassis Distribution Switches (dist-sw-01, dist-sw-02) with ECMP hashing.

Access & Workstations: 4x Access Switches feeding 8x departmental endpoints (IT, Engineering, Finance, DMZ).

ENTERPRISE CIDR SUBNET SEGREGATION
• 10.0.1.0/24 • VLAN 10 • IT & Management Network
• 10.0.2.0/24 • VLAN 20 • Engineering & DevOps Workstations
• 10.0.3.0/24 • VLAN 30 • Finance & Payroll Segment
• 10.0.10.0/24 • VLAN 100 • DMZ App & DB Server Cluster

8. CHAOS & FAST RECOVERY

SUB-SECOND

Built-in chaos engineering APIs and UI context menus allow operators to inject faults and verify self-healing recovery in real-time:

Right-Click Link Menu: Right-click any cable on the canvas to set Link Down, Impair (loss/jitter), or Trace.

QoS Impairment: Injects synthetic 40% loss and +35ms latency to model optical degradation or dirty fiber.

Fast Metric Reset: Upon physical link restoration, DNMS flushes stale route failures immediately.

VERIFIED SUB-SECOND CONVERGENCE BENCHMARK
• **T = 0.0s (Fault):** Link severed → Traversing paths fail ACKs.
• **T = 2.5s (Alert):** Suspicion spikes to **+92.4%** → Red **FAULT SUSPECT** badge appears.
• **T = 4.0s (Restore):** Operator clicks 'Restore'.
• **T = 4.8s (Recovery):** ResetRoutesForLink flushes ring buffers → **Suspicion collapses back to 0.0% within 1.0 second!**

9. TELEMETRY BENCHMARK

COMPARISON

COMPETITIVE ARCHITECTURE BENCHMARK

Metric	Legacy SNMP	DNMS Platform
Detection Time	60–300s	**< 3.0s (Real-time)**
ECMP Visibility	Blind	**100% Swept Multi-Path**
Credentials	SSH/SNMP Required	**Zero Credentials**
Overhead	Heavy MIB polling	**< 0.01% Link Bandwidth**
Packet Inspection	None (CLI only)	**Live Wireshark Dock**
Cluster Model	Central Server	**Decentralized SWIM Mesh**
Recovery Latency	Minutes	**< 1.0s Ring Flush**

1-Click Verification: .\start_cluster.ps1
Prometheus scrape endpoint: http://localhost:12345/metrics